

The Strong-CP Boundary Endpoint

An AASC Constraint Theorem for CP-Odd QCD Topological Standing

Amos Jay Maley

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Abstract

This paper treats the strong-CP smallness problem as a boundary-admissibility and CP-odd topological-standing problem. The Admissibility And Standing Constraint framework is used as a mathematical constraint formalism, not as an additional QCD model, an axion substitute, a naturalness prior, or an interpretive overlay. Standard QCD supplies the comparison arena: the topological θ -term, anomalous chiral transport, quark-mass determinant phase, topological-sector weighting, and the physical quotient survivor

$$\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}.$$

AASC determines which promotions, boundary loads, selectors, branch claims, and endpoints retain standing in that arena. The dependency direction is kernel-first: any determinate same-target strong-CP endpoint evaluation already requires reference, standing, admissibility, and irreversibility before any candidate value can claim endpoint role.

The first result is the strong-CP quotient ledger: chiral rotations, anomaly/Jacobian book-keeping, mass-basis changes, CKM rephasings, determinant branches, vacuum relabelings, and scale/scheme transport preserve $\bar{\theta}$ or change target. The second result is the QCD-to-boundary bridge: because $G\tilde{G}$ is locally a total derivative but globally weights topological sectors, the quotient survivor $\bar{\theta}$, not the displayed bare angle alone, is the CP-odd topological boundary class subject to standing audit. Generic nonzero classes are exchanged with CP-conjugate sector-weighting classes, while every nonzero class, including the sign-fixed π class, is non-identity topological load. The third result is the selector-burden theorem: nonzero QCD topological boundary load can have primitive fixed-scope standing only if a certified selector package, declared boundary input, positive empirical CP-odd fixing witness, or richer-scope descent supplies the missing standing-bearing source. A displayed θ -term or representative θ -coordinate, a chosen chiral basis, a determinant branch, an anomaly equation, a regulator convention, a θ -vacuum label, or an EDM upper bound does not satisfy that burden.

The endpoint is AASC-internal in the precise necessity sense: any non-degenerate regime of claim-bearing endpoint use for the fixed strong-CP target already instantiates the kernel roles that the AASC consequence layer audits. In $D_{\text{SCP},0}$, no same-scope selector package survives exhaustion. Writing $C_\kappa := \{\bar{\theta} = \kappa\}$ for the family of candidate quotient classes, every non-identity class is deleted from the certified primitive survivor set, while C_0 is the identity class carrying no nonzero topological boundary load. The selected fixed-scope endpoint is

$$\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1, \quad C_0 = \{\bar{\theta} = 0\}.$$

A nonzero parameterized theory $\text{QCD}_{\bar{\theta}=\epsilon}$ is formally writable, but $\epsilon \neq 0$ is a declared boundary insertion, witness-driven update, or richer-scope descent rather than an internally selected primitive endpoint. Peccei-Quinn/axion, UV CP/parity, Nelson-Barr, flavor/modular, massless-quark, and future nonzero EDM routes are not objections to the theorem; they are exactly the selector, input, or target-update packages the theorem says would be required.

In the usual physics idiom, the result says: without new physics, the fixed strong-CP target admits no primitive nonzero $\bar{\theta}$ endpoint. More exactly, absent a certified selector, declared nonzero boundary input, positive empirical fixing witness, or richer-scope source-continuity package, the only candidate class that carries primitive endpoint standing is $C_0 = \{\bar{\theta} = 0\}$.

Keywords: strong CP problem; QCD θ -term; boundary admissibility; standing; no-promotion; θ -vacuum; anomaly; axion; Peccei-Quinn; Nelson-Barr; AASC.

Mathematical-status lock and proof-class boundary

AASC is used here as a mathematical constraint formalism, not as philosophical commentary, metaphor, optional narration, or a phenomenological substitute for QCD. The proof is the theorem chain stated in this manuscript. The AASC primitives used below—reference, standing, admissibility, irreversibility, boundary exactness, no-promotion, and no hidden selector discipline—are fixed-domain constraints for claim-bearing primitive endpoint use. Standard QCD supplies the comparison arena; AASC supplies the status calculus by which formal availability, primitive endpoint standing, selector import, declared input, and target update are separated.

A same-target objection must therefore identify a failure in a definition, theorem link, dependency claim, QCD-to-boundary-load bridge, selector-burden theorem, transport normal form, endpoint-claim audit, or final selection inference. It is not a refutation merely to classify the vocabulary as unfamiliar, to demand an axion model, a lattice calculation, a UV completion, a naturalness prior, or a phenomenological forecast, or to observe that standard QCD formally contains a topological coordinate slot. Those are different theorem classes.

Read this first: strong-CP endpoint proof spine

The proof has three layers. The first layer fixes the admissibility conditions under which a strong-CP endpoint claim can be evaluated at all. The second layer identifies the QCD quotient object and its candidate endpoint classes. The third layer exhausts the nonzero candidates and selects the quotient identity.

Layer I: kernel-forced endpoint use. A determinate claim-bearing endpoint use of the fixed strong-CP target already requires the kernel roles:

$$\text{TargetAdeq}_{D_{\text{SCP}}} \Rightarrow K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D).$$

Thus AASC is not an optional overlay on QCD. It names the consequence layer already forced by fixed-target claim-bearing endpoint evaluation.

Layer II: strong-CP quotient and candidate classes. Standard QCD supplies the topological coordinate slot, anomalous chiral transport, and quark-mass determinant phase. The standing-relevant quotient survivor is

$$\Theta_{\text{slot}} = \mathbb{R}/2\pi\mathbb{Z}, \quad \bar{\theta} = [\theta + \arg \det M_q]_{2\pi}.$$

The endpoint candidates are the quotient classes

$$C_\kappa := \{\bar{\theta} = \kappa\}, \quad \kappa \in \mathbb{R}/2\pi\mathbb{Z}.$$

The identity candidate is

$$C_0 := \{\bar{\theta} = 0\}.$$

Layer III: nonzero-candidate deletion and identity selection. For every nonzero candidate,

$$\kappa \neq 0 \Rightarrow \text{FreeTopPhase}_{D_{\text{SCP},0}}(C_\kappa) \wedge \text{NonIdentityTopologicalLoad}_{D_{\text{SCP},0}}(C_\kappa).$$

A nonzero freely variable non-identity topological boundary-load class can survive only if it has a certified source:

$$\text{SelectorPkg}_{\bar{\theta}}^{\text{cert}}(C_\kappa) \vee \text{Input}_{D_{\text{SCP},0}}(C_\kappa) \vee \text{WitFix}_{D_{\text{SCP},0}}(C_\kappa) \vee \text{Rich}_{D_{\text{SCP},0}}(C_\kappa).$$

In ordinary physics language, each disjunct is a concrete source-continuity package. A vague appeal to an unknown stabilizer for a small nonzero $\bar{\theta}$ does not instantiate any disjunct. If the stabilizer is real, it must be specified as new fields, new symmetries, boundary conditions, determinant-protection structure, dynamical relaxation, a positive fixing witness, or another certified source-continuity mechanism. Once specified, it updates the target to $D_{\text{SCP},+}$.

In $D_{\text{SCP},0}$, no such source is present. Therefore the nonzero family is deleted uniformly:

$$\forall \kappa \neq 0, \quad \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0.$$

The identity class is available and primitive-standing because it is the quotient identity, not a chosen coefficient-zero branch:

$$\text{Avail}_{D_{\text{SCP},0}}(C_0) = 1, \quad \text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_0) = 1.$$

Hence the fixed strong-CP endpoint is

$$\boxed{\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1, \quad C_0 = \{\bar{\theta} = 0\}.$$

No one-step retyping. The proof does not use any shortcut of the form

formal θ -writability $\Rightarrow$ primitive endpoint standing,

$$\neg \text{SelectorPkg}_{\bar{\theta}}^{\text{cert}}(C_\kappa) \Rightarrow \bar{\theta} = 0 \quad \text{without boundary-load classification,}$$

or

$$\theta = 0 \Rightarrow C_0 \quad \text{as a chosen coefficient-zero branch.}$$

The typed route is

$$\begin{aligned} \Theta_{\text{slot}} &\rightarrow \bar{\theta} \rightarrow C_\kappa \\ &\rightarrow \text{FreeTopPhase} \rightarrow \text{NonIdentityTopologicalLoad} \\ &\rightarrow \text{selector burden} \rightarrow \text{family-wise deletion} \\ &\rightarrow \text{identity selection.} \end{aligned}$$

The theorem therefore selects C_0 only after the quotient object is fixed, every nonzero candidate is classified, every same-scope transport and endpoint-claim route is exhausted, and the identity class is certified under the selection criterion.

Dependency-inversion exclusion

The dependency order is not

AASC framework $\Rightarrow$ strong-CP constraint.

The dependency order is

determinate fixed-scope strong-CP endpoint evaluation
 $\Rightarrow K_D \Rightarrow$ boundary-standing and selector-burden consequences.

A formally writable θ -coordinate slot cannot define the standing rule by which a nonzero $\bar{\theta}$ class becomes primitive endpoint standing. A same-target evaluation of $\bar{\theta}$ already requires a fixed quotient object, a candidate family, lawful same-scope transport, branch status, endpoint standing, and act-time finality. AASC names and audits the consequence layer of those forced roles; it is not a discretionary second standard placed on an otherwise complete endpoint act.

Contents

Mathematical-status lock and proof-class boundary	2
Read this first: strong-CP endpoint proof spine	2
Dependency-inversion exclusion	4
Executive result and claim classes	7
1 Introduction	9
1.1 Proof class in the strong-CP target	9
1.2 The constraint target	9
1.3 AASC as a constraint formalism	10
1.4 Dependency direction and dependency-inversion exclusion	10
1.5 Targets that must remain separate	11
1.6 Standard inputs and AASC outputs	11
1.7 Main results	12
1.8 Nonclaims and scope boundaries	12
2 Constraint kernel, target lock, and status calculus	13
2.1 Kernel necessity and boundary standing	13
2.2 Boundary non-promotion	15
2.3 Source lock	15

3	QCD strong-CP ledger	16
3.1	The θ -term and topological charge	16
3.2	Quark mass phase and quotient survivor	16
3.3	Weak CP firewall	17
3.4	Empirical interface	17
4	From theta to CP-odd boundary load	17
4.1	Topological-sector bridge	17
4.2	Availability is not occupancy	18
4.3	Parameter-status taxonomy	18
5	Same-scope transport and endpoint-claim audit	19
5.1	Audit principle and normal forms	19
5.2	Mini-lemmas	20
6	Selector burden and CP-odd boundary no-promotion	23
6.1	Boundary source continuity	23
6.2	Non-identity topological load	24
6.3	Selector-burden theorem	25
6.4	Primitive CP-odd boundary load impossibility	25
6.5	Identity-class certification	25
6.6	Fixed-scope strong-CP endpoint	26
7	Declared input, $\theta = \pi$, and parameterized QCD	27
7.1	Declared-input compatibility	27
7.2	The $\theta = \pi$ class	27
8	Richer-scope selector normal form	28
8.1	General certificate	28
8.2	Peccei-Quinn and axion relaxation	28
8.3	UV CP, parity, Nelson-Barr, and flavor routes	29
8.4	Empirical witness update	29
9	Same-target closure and counterexample discipline	29

9.1	AASC dependence as theorem class	29
9.2	Cost of kernel denial	29
9.3	No strict same-target weakening	30
9.4	Faithful same-regime counterexample conditions	31
9.5	Right-mode objection targets	31
9.6	Standing-objection register	32
10	Conclusion	33
A	Failure-route table	34
B	Claim and nonclaim register	35
C	Forbidden-promotion register	36
D	Anti-circularity audit	37
E	Theorem-level source dependency map	38
F	Notation and glossary	39

Executive result and claim classes

The proof spine is

$$\begin{aligned}
\text{TargetAdeq}_{D_{\text{SCP}}} &\Rightarrow K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D) \\
&\Rightarrow \text{BoundaryStand}_D \Rightarrow \bar{\theta} \in \mathbb{R}/2\pi\mathbb{Z} \\
&\Rightarrow \forall \kappa \neq 0, \text{NonIdentityTopologicalLoad}_{D_{\text{SCP},0}}(C_\kappa) \Rightarrow \text{SelectorPkg}_{\bar{\theta}}^{\text{cert}} \text{burden} \quad (1) \\
&\Rightarrow \forall \kappa \neq 0, \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0 \Rightarrow \text{Avail}_{D_{\text{SCP},0}}(C_0) = 1 \\
&\Rightarrow \text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_0) = 1 \Rightarrow \text{Sel}_{D_{\text{SCP},0}}(C_0) = 1.
\end{aligned}$$

The arrows are status dependencies, not claims that this paper rederives the whole realized-physics program. The kernel, structure-of-admissibility, endpoint-exactness, anomaly, holonomy, mass-matrix, vacuum, and fixed-domain exhaustion sources supply the AASC consequence layer. Standard QCD identities define the comparison arena.

Definition 0.1 (Fixed strong-CP target). $D_{\text{SCP},0}$ is the fixed AASC strong-CP endpoint target consisting of:

- (i) the ordinary QCD/SM strong-sector gauge and quark-mass carrier inventory;
- (ii) nontrivial QCD topological sectors and the topological coordinate slot $\Theta_{\text{slot}} := \mathbb{R}/2\pi\mathbb{Z}$; the target contains the slot and quotient object, not a declared nonzero value;
- (iii) anomalous chiral transport and the determinant phase quotient;
- (iv) the weak-sector CP ledger as a separate ledger, not a strong-sector selector;
- (v) no axion/PQ carrier, no UV CP/parity selector, no Nelson-Barr or determinant-protection extension, no certified flavor/modular selector, no declared nonzero strong-CP boundary input, and no positive empirical strong-sector CP-odd fixing witness.

The update target $D_{\text{SCP},+}$ is triggered by a certified selector package, declared nonzero input, positive fixing witness, or richer-scope descent. A merely unspecified possible mechanism does not trigger $D_{\text{SCP},+}$. The update requires a specified source-continuity package: a concrete carrier, law, symmetry, boundary condition, dynamical coordinate, empirical fixing record, or descent map capable of carrying the nonzero class. Such an update is not a counterexample to the theorem in $D_{\text{SCP},0}$. The definition of $D_{\text{SCP},0}$ does not define $\bar{\theta} \neq 0$ away. It fixes the target in which nonzero primitive boundary standing is tested. Nonzero $\bar{\theta}$ is deleted only after the topological-load, selector-burden, source-continuity, and transport/endpoint-claim exhaustion theorems are applied.

Proposition 0.2 (Context non-smuggling). The fixed target $D_{\text{SCP},0}$ fixes the strong-CP carrier, quotient object, coordinate slot, and candidate family only. It does not assert

$$\text{Sel}_{D_{\text{SCP},0}}(C_0), \quad \text{Surv}_{D_{\text{SCP},0}}(C_0), \quad \neg \text{Surv}_{D_{\text{SCP},0}}(C_\kappa), \quad \bar{\theta} = 0.$$

The deletion of nonzero candidates occurs only after the QCD quotient bridge, the free-topological-phase classification, the non-identity load theorem, the selector-burden theorem, and the endpoint-claim audit have been applied.

Definition 0.3 (Theta-coordinate slot and candidate values). *The fixed target $D_{\text{SCP},0}$ contains the strong topological coordinate slot*

$$\Theta_{\text{slot}} := \mathbb{R}/2\pi\mathbb{Z}, \quad (2)$$

not a declared nonzero coordinate value. Candidate values are audited only after the chiral quotient has been formed: $C_\kappa = \{\bar{\theta} = \kappa\}$. Thus $D_{\text{SCP},0}$ contains the object-space in which strong-CP boundary standing is tested, while a stipulated nonzero value $\kappa \neq 0$ is a declared boundary insertion or target update unless it is independently selected, witnessed, or inherited.

Remark 0.4 (Declared-input preview). *A parameterized theory $\text{QCD}_{\bar{\theta}=\epsilon}$ remains formally writable. For $\epsilon \neq 0$, it is not a primitive endpoint of $D_{\text{SCP},0}$; it is a declared boundary-input target, positive-witness target, or richer-scope update. This does not assert that nonzero $\bar{\theta}$ is impossible in nature. It asserts that a nonzero value requires a specified source-continuity route and therefore is not primitive same-target standing in $D_{\text{SCP},0}$. The compatibility theorem in Section 7 records this status classification and does not weaken the endpoint theorem in $D_{\text{SCP},0}$.*

Definition 0.5 (Candidate strong-CP quotient classes). *For each $\kappa \in \mathbb{R}/2\pi\mathbb{Z}$, write*

$$C_\kappa := \{\bar{\theta} = \kappa\} \quad (3)$$

for the candidate quotient-endpoint class. The phrase $\bar{\theta} \neq 0$ abbreviates the family $\{C_\kappa : \kappa \in \mathbb{R}/2\pi\mathbb{Z}, \kappa \neq 0\}$. Formal endpoint selection is stated over the classes C_κ , with C_0 denoting the identity class.

Definition 0.6 (Positive fixing witness and bound record). *$\text{WitFix}_D(C_\kappa)$ denotes a positive or fixing empirical strong-sector CP record that supports a nonzero class, selects a boundary class, or forces a target update. $\text{CPBound}_D(\bar{\theta})$ denotes an upper-bound or null-channel constraint on strong CP. These statuses are distinct:*

$$\text{CPBound}_D(\bar{\theta}) \not\Rightarrow \text{Sel}_D(C_0), \quad \text{CPBound}_D(\bar{\theta}) \not\Rightarrow \text{WitFix}_D(C_\kappa). \quad (4)$$

An upper bound constrains admissible reconstructions; it is not a positive fixing witness and not an exact endpoint selector.

Definition 0.7 (Freely variable strong-CP topological phase). *A candidate topological endpoint class C_κ is freely variable in a target D , written $\text{FreeTopPhase}_D(C_\kappa) = 1$, when no gauge-consistency theorem, anomaly-matching theorem, quantization law, declared boundary law, selector package, positive fixing witness, or richer-scope descent fixes κ . Placeholder language such as “some unknown stabilization mechanism” does not remove free variability unless it identifies the carrier, law, boundary condition, witness, or descent map by which κ is fixed. In $D_{\text{SCP},0}$, every nonzero strong-CP candidate class C_κ is freely variable until such a source is supplied.*

Definition 0.8 (Strong-CP selected endpoint). *The selected fixed-scope endpoint in $D_{\text{SCP},0}$ is the unique certified same-target endpoint class after all same-scope transports, redescription skins, hidden selector routes, calibration imports, and richer-scope exits are deleted or quotient-collapsed.*

Table 1: Claim classes used in this paper.

Class	Meaning
Imported AASC role theorem	A kernel, boundary, exactness, anomaly, holonomy, vacuum, mass-matrix, or fixed-domain result fixing the standing discipline used in the strong-CP specialization.
Standard QCD input	A conventional field-theory identity used to define the comparison arena: the θ -term, topological charge, anomalous axial shift, quark mass phase, or EDM interface.
Constraint-exhaustion theorem	A deletion, no-promotion, quotient-collapse, or burden theorem obtained by exhausting all declared same-target routes.
Endpoint theorem	AASC selection over certified same-target survivors inside $D_{\text{SCP},0}$.
Scope boundary	A stronger or different claim intentionally excluded because it changes the target, imports new carrier load, or requires a separate certificate.

1 Introduction

1.1 Proof class in the strong-CP target

The front matter fixes the proof status: AASC is the mathematical constraint formalism and standard field theory defines the comparison arena. This section fixes the strong-CP target to which that proof class is applied. The theorem is a kernel-forced boundary-admissibility endpoint theorem, not a phenomenological fit, model preference, chiral-basis maneuver, or request that standard QCD adopt AASC language before the theorem can be stated.

The P=NP endpoint manuscript is cited only as an example of AASC endpoint-posture architecture–mathematical-status lock, dependency-inversion exclusion, cost-of-denial framing, and right-mode objection discipline. No P=NP endpoint conclusion is imported. The role of the cited posture is to make explicit that standard comparison inputs are not advertised as new physics; the contribution here is the strong-CP role, selector, promotion, and endpoint calculus [38].

1.2 The constraint target

The conventional strong-CP problem begins with the QCD topological term

$$\mathcal{L}_\theta = \frac{g_s^2}{32\pi^2} \theta G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (5)$$

In a theory with massive quarks, the physical strong-sector invariant is not θ alone. The invariant quotient is

$$\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}, \quad (6)$$

up to sign convention. Chiral rephasings move phase between the displayed topological term and the quark mass determinant. They do not remove $\bar{\theta}$.

The target of this paper is not another strong-CP model. It is the standing of the promotion

formally writable CP-odd topological coordinate slot $\longrightarrow$ primitive fixed-scope endpoint.

The operative question is:

Given the fixed strong-CP target $D_{\text{SCP},0}$, can nonzero $\bar{\theta}$ remain a certified primitive same-target endpoint without a certified selector package, declared boundary input, positive empirical fixing witness, or richer-scope descent?

The answer is no. The selected endpoint in $D_{\text{SCP},0}$ is $\bar{\theta} = 0$.

1.3 AASC as a constraint formalism

AASC's native proof modes are exhaustion, impossibility, deletion, quotient collapse, burden transfer, and no-promotion. It does not add particles, symmetries, or interactions. It removes alternatives that cannot preserve the declared target without hidden load, unlicensed repair, outcome-conditioned selection, calibration import, or scope change.

A negative conclusion is not an incomplete positive model. It is the theorem output when a proposed route cannot perform the role it claims while preserving the comparison object. Here the route being tested is the promotion of nonzero CP-odd topological boundary load to primitive fixed-scope standing.

1.4 Dependency direction and dependency-inversion exclusion

The dependency direction is

$$\text{determinate strong-CP boundary evaluation} \Rightarrow K_D \Rightarrow \text{AASC consequence layer}. \quad (7)$$

It is not

$$\text{a formally writable } \theta\text{-coordinate slot} \Rightarrow \text{the standing rule by which theta wins}.$$

A candidate nonzero boundary parameter cannot define the selector by which it is promoted. Once the target is a determinate same-scope CP-odd topological boundary evaluation, the kernel roles are already in force: reference fixes the target, standing makes successful content reusable, admissibility separates licensed continuation from failure, and irreversibility blocks post-failure repair as the same act [19, 20].

Theorem 1.1 (Dependency-inversion exclusion). *A proposed kernel-independent construction that retains the same determinate strong-CP endpoint target reverses dependency order. It can survive only by showing that one of the target-adequacy roles is unnecessary, deriving the corresponding kernel work from a strictly lower non-circular basis, or realizing a faithful same-regime counterexample preserving $D_{\text{SCP},0}$ while certifying nonzero primitive CP-odd boundary standing without selector, input, witness, or richer descent.*

Proof. Target determinacy, branch evaluability, act-time finality, and same-regime fidelity are required before $\bar{\theta} = 0$, $\bar{\theta} \neq 0$, declared input, axion descent, or CP/parity protection can be compared as claims about the same object. Granting those roles grants the governance work of the kernel under whatever vocabulary is used. Denying one of them loses the corresponding fixed-target good. Reintroducing the lost good through a hidden selector, branch convention, empirical retrofit, or target translation imports governance-equivalent work or changes scope. $\square$

1.5 Targets that must remain separate

The phrase “solution of the strong-CP problem” merges several targets:

1. the quotient identity of the physical invariant $\bar{\theta}$;
2. the empirical smallness bound inferred from neutron EDM limits;
3. the status of $\text{QCD}_{\bar{\theta}=\epsilon}$ as a parameterized family;
4. primitive fixed-scope standing of a nonzero CP-odd boundary class;
5. a dynamic relaxation model such as PQ/axion;
6. a protective UV CP/parity or determinant route;
7. a future empirical nonzero CP-odd strong-sector witness.

This paper proves the fourth target relative to the first. The invariant $\bar{\theta}$ is the object; primitive nonzero endpoint standing is the branch under audit.

1.6 Standard inputs and AASC outputs

Table 2: Standard QCD inputs versus AASC constraint outputs.

Standard comparison input	AASC constraint output
$G\tilde{G}$ is formally writable in the QCD action.	Formal writability is not primitive boundary standing.
$G\tilde{G} = \partial_\mu K^\mu$ locally but integrates to topological charge globally.	The θ angle is a topological-sector weighting datum subject to boundary audit.
Chiral rotations move phase between θ and $\arg \det M_q$.	$\bar{\theta}$ is the quotient survivor; redescription is not erasure.
Generic nonzero $\bar{\theta}$ classes are exchanged with CP-conjugate sector-weighting classes; the sign-fixed π class is nonzero but non-identity.	Generic nonzero classes carry CP-odd boundary load; every nonzero class, including C_π , is non-identity topological boundary load.
$\text{QCD}_{\bar{\theta}=\epsilon}$ is a parameterized formal family.	$\epsilon \neq 0$ is declared boundary input unless selected, witnessed, or inherited.

Standard comparison input	AASC constraint output
EDM bounds constrain $\bar{\theta}$.	Empirical bounds are witness constraints, not internal endpoint selection.
PQ/axion can relax effective $\bar{\theta}$.	PQ/axion is a richer-scope dynamic selector.
UV CP/parity or Nelson-Barr can protect $\bar{\theta}$.	Such routes are richer-scope protective selector packages.

1.7 Main results

The paper proves six levels of result.

- (i) **Quotient-invariant spine.** In fixed QCD/SM with massive quarks, $\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}$ is the unique strong-sector CP-odd quotient survivor.
- (ii) **QCD-to-boundary bridge.** The survivor $\bar{\theta}$ weights topological sectors; generic nonzero classes are exchanged with CP-conjugate sector-weighting classes, and every nonzero class, including the sign-fixed π class, is non-identity topological load.
- (iii) **Selector-burden theorem.** Nonzero QCD topological boundary load requires a certified selector package, declared boundary input, positive empirical fixing witness, or richer-scope descent.
- (iv) **Transport and endpoint-claim exhaustion.** Chiral, anomaly, determinant, CKM, vacuum, holonomy, and renormalization transports preserve the quotient or fail to select it; empirical upper bounds, declared values, richer-scope mechanisms, and determinant-rank routes are endpoint-claim statuses rather than same-scope selectors.
- (v) **Reopening condition.** A nonzero C_κ can re-enter only through a specified source-continuity package: new fields, symmetries, determinant-protection structure, boundary conditions, dynamical relaxation, declared input, or positive empirical fixing witness. A vague unknown stabilizer is not a certified selector and does not reopen the fixed target.
- (vi) **Endpoint collapse.** In $D_{\text{SCP},0}$, nonzero $\bar{\theta}$ has no certified primitive survivor; the identity class $\bar{\theta} = 0$ is selected.

1.8 Nonclaims and scope boundaries

- This is not a conventional-QFT-only dynamical minimization theorem.
- This is not a proof that nonzero-theta QCD cannot be formally written, simulated, or studied.
- This is not a claim that every SM parameter must be predicted by AASC.
- This is not an anti-axion theorem. Axions are classified as legitimate richer-scope selectors when their certificate closes.
- This is not a claim that nonzero $\bar{\theta}$ is impossible in nature. A nonzero value would require specified source continuity and therefore a target update.

- This is not a uniqueness theorem for UV origin. The endpoint is fixed-scope and primitive-boundary-status specific.

2 Constraint kernel, target lock, and status calculus

2.1 Kernel necessity and boundary standing

The kernel dependency is

$$\text{TargetAdeq}_D \Rightarrow K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D). \quad (8)$$

Reference fixes what is being evaluated; standing makes successful content reusable; admissibility separates licensed continuation from failure; irreversibility prevents a failed act from being repaired as the same act. Boundary exactness then blocks a boundary trace from becoming endpoint authority merely because it is writable or useful in a calculation [19, 20, 25].

Definition 2.1 (Kernel-neutral strong-CP target adequacy). *A strong-CP endpoint evaluation is target-adequate when it satisfies four neutral conditions.*

- (A1) Target determinacy: *the object of evaluation is fixed as the quotient family $C_\kappa = \{\bar{\theta} = \kappa\}$ inside $\mathbb{R}/2\pi\mathbb{Z}$, rather than drifting among θ , $\arg \det M_q$, $\bar{\theta}$, an EDM constraint, an axion relaxation coordinate, or a declared parameter family.*
- (A2) Status evaluability: *formal writability, quotient survival, primitive endpoint standing, declared boundary input, positive fixing witness, upper-bound/null-channel constraint, and richer-scope descent are distinguishable statuses.*
- (A3) Act-time finality: *a failed nonzero candidate is not repaired later by importing PQ/axion, UV CP/parity, determinant protection, CKM descent, declared input, or empirical witness while being treated as the same fixed-scope endpoint act.*
- (A4) Same-regime fidelity: *lawful redescription preserves the same QCD/SM carrier, quotient object, candidate family, transport laws, and endpoint-status classification.*

Definition 2.2 (Claim-bearing primitive endpoint use). $\text{EndpointUse}_{D_{\text{SCP},0}}(C_\kappa)$ means that the candidate class C_κ is used as a claim-bearing primitive endpoint of the fixed strong-CP target. The theorem-bearing act is the manuscript's endpoint-selection claim; the bare QCD coordinate value is only a candidate class until standing, selector, input, witness, or target-update status is certified.

Theorem 2.3 (Strong-CP endpoint use is governed construction). *For any candidate class C_κ , claim-bearing primitive endpoint use in $D_{\text{SCP},0}$ instantiates the kernel:*

$$\text{EndpointUse}_{D_{\text{SCP},0}}(C_\kappa) \Rightarrow \text{GovConstr}_{D_{\text{SCP},0}}(C_\kappa) \Rightarrow K_D. \quad (9)$$

Consequently, fixed-scope endpoint claims about C_κ are subject to boundary-standing, no-promotion, selector-burden, and same-scope transport discipline.

Proof. Claim-bearing primitive endpoint use fixes the strong-CP target as a quotient candidate class, so target determinacy supplies reference. It distinguishes the status of a writable term,

quotient survivor, primitive endpoint, declared input, witness update, and richer descent, so status evaluability supplies standing. It does not permit later selector import to repair the same failed endpoint act, so act-time finality supplies irreversibility. It preserves the same carrier and quotient under lawful redescription, so same-regime fidelity supplies the admissibility boundary. These are exactly the kernel roles. The downstream AASC machinery is therefore the consequence layer of claim-bearing endpoint use, not a separate source of standing. $\square$

Consequently, once a regime commits to determinate target identity, evaluable endpoint-claim status, act-time finality, and identity-preserving redescription, the kernel roles are no longer optional. Reference is the fixed target object; standing is the evaluability of the claim as more than a written coefficient; admissibility is the boundary between lawful same-scope continuation and scope-changing import; irreversibility prevents a failed endpoint act from being repaired later as the same act. Any attempted weakening that preserves the strong-CP target must either reintroduce the same governance work under different vocabulary or fall into one of the documented failure modes: target drift, status collapse, repair, hidden selector import, or domain change. The AASC machinery used below therefore records consequences already forced by the target, not an external standard imposed afterward.

Definition 2.4 (Fixed-domain consequence layer). $A_{D_{\text{SCP},0}}^+(C_\kappa)$ denotes the endpoint consequence layer forced by kernel-instantiated claim-bearing strong-CP endpoint use: boundary fixation, standing as reuse-stable admissibility, no hidden selector, no same-act repair, no third endpoint status, no primitive boundary-load promotion without source-continuity, no report beyond carrier support, transport closure, and endpoint-selection discipline.

Proposition 2.5 (Downstream status of AASC machinery). *Boundary-standing discipline, selector-burden discipline, no-promotion, same-scope route exhaustion, endpoint-claim audit, and identity-class selection are downstream consequence machinery forced by the kernel. They are not primitive sources of theorem standing.*

Proof. By Theorem 2.3, claim-bearing endpoint use already instantiates reference, standing, admissibility, and irreversibility. The later AASC machinery preserves those roles under boundary, transport, quotient, selector, and report operations. Hence it records consequences of the kernel rather than adding an independent source of authority. $\square$

Proposition 2.6 (Proof calculus and non-tautological endpoint content). *The endpoint theorem is proof-calculus dependent in the ordinary sense that every formal theorem is evaluated inside a declared proof calculus. It is not tautological. The definitions fix the admissibility target, candidate family, selector burden, and status vocabulary. They do not by themselves decide the strong-CP endpoint.*

Proof. The endpoint follows only after the QCD quotient survivor $\bar{\theta}$ is identified, the candidate classes C_κ are formed, every nonzero C_κ is classified as freely variable non-identity topological boundary load, same-scope transports are exhausted, endpoint-claim routes are separated from selectors, every nonzero survivor is deleted uniformly, and the quotient identity C_0 is certified as available and primitive-standing. A same-target counterexample must therefore preserve $D_{\text{SCP},0}$ while producing primitive standing for some $C_\kappa \neq C_0$ without selector, declared input, positive fixing witness, or richer-scope descent. That is a theorem-level burden, not a restatement of the definition. $\square$

Definition 2.7 (Formal availability). $\text{Avail}_D(C) = 1$ means that candidate class C can be written in the declared field, gauge, locality, Lorentz, quotient, and boundary syntax of D . Availability alone is not occupancy, selection, or primitive standing.

Definition 2.8 (Primitive standing). $\text{PrimitiveStanding}_D(C) = 1$ means that candidate class C has fixed-domain standing without relying on an undeclared boundary input, hidden selector, post-hoc branch, empirical calibration import, or richer-scope carrier.

Definition 2.9 (Certified selector package). A nonzero strong-CP selector package is

$$\text{SelectorPkg}_\theta^{\text{cert}} = \text{Carrier}^{\text{cert}} \wedge \text{SelectionLaw}^{\text{cert}} \wedge \text{TransportStability}^{\text{cert}} \wedge \text{LowEnergyDescent}^{\text{cert}}. \quad (10)$$

Depending on route, the carrier may be an axion field, a PQ symmetry, a UV CP/parity rule, a determinant-protection structure, a flavor/modular carrier, a declared boundary condition, or a positive strong-sector CP-odd fixing record. A written coefficient, representative branch, anomaly equation, regulator convention, vacuum label, or null bound is not this package.

Criterion 2.10 (Standing-preserving endpoint selection). For a fixed target D and quotient relation $\sim_D$,

$$\text{Sel}_D(C) = 1 \iff \text{Avail}_D(C) = 1 \wedge \text{PrimitiveStanding}_D(C) = 1 \wedge \forall C' \not\sim_D C, \text{Surv}_D^{\text{cert}}(C') = 0. \quad (11)$$

Selection is uniqueness over certified primitive survivor quotients, not over writable classes.

2.2 Boundary non-promotion

Principle 2.11 (Boundary non-promotion). A boundary datum does not acquire primitive endpoint standing merely because a calculation can be performed relative to it. If it is non-neutral boundary load and no selector, declared input, positive empirical fixing witness, or richer-scope descent carries it, then it is boundary-relative input, hidden selector, presentation skin, or scope extension, not primitive standing.

This principle is the strong-CP specialization of the endpoint and boundary exactness arc: endpoint content must be target-declared, redescription-stable, calibration-free, channel-preserving, non-repaired, and not introduced as a hidden branch selector [21–26].

2.3 Source lock

Table 3: Focused source lock and permitted reuse.

Source family	Imported role and reuse boundary
Kernel and structure of admissibility	Target adequacy forces reference, standing, admissibility, and irreversibility; AMetric pre-boundary structure supplies no arbitrary selector.
Endpoint and exactness arc	Boundary traces, quotient representatives, calibration imports, and transport artifacts cannot promote themselves to raw endpoint exactness.

Source family	Imported role and reuse boundary
Structura and Standard-Model structure sources	The strong-CP invariant is $\bar{\theta}$; the full QCD/SM comparison arena is fixed before branch evaluation.
Anomaly ledger sources	Anomaly/Jacobian, determinant, measure, and current equations account for shifts; they do not select values.
Holonomy and CP-like phase sources	Phase, residue, loop, holonomy, and CP-like classes have standing only after quotient and witness discipline.
Mass-matrix and mixing-phase sources	$\arg \det M_q$ and CKM phases remain typed; determinant branch and weak CP phase cannot silently select strong CP.
Gauge-vacuum and residual-sector sources	Vacuum representative choices, residual labels, and θ -vacuum relabeling do not erase the quotient survivor.
Fixed-domain transport and endpoint-claim exhaustion sources	Same-target alternatives must break a load-bearing link or supply the missing certificate.

3 QCD strong-CP ledger

3.1 The θ -term and topological charge

The QCD θ -term is Eq. (5). Locally,

$$\frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu} = \partial_\mu K^\mu, \quad (12)$$

where K^μ is the Chern-Simons current. Globally, for finite-action configurations with the usual boundary conditions,

$$Q_{\text{top}} = \frac{g_s^2}{32\pi^2} \int d^4x G_{\mu\nu}^a \tilde{G}^{a\mu\nu} \in \mathbb{Z}. \quad (13)$$

The Euclidean path-integral sectors are weighted by $e^{i\theta Q_{\text{top}}}$. This is the standard reason θ is not merely a local perturbative coefficient. It is a topological-sector weighting datum. The instanton, θ -vacuum, anomalous-current, and topological-charge facts used here are standard field-theory inputs to the AASC audit rather than AASC-derived identities [1–5].

3.2 Quark mass phase and quotient survivor

The quark mass term is

$$\mathcal{L}_m = -\bar{q}_R M_q q_L - \bar{q}_L M_q^\dagger q_R. \quad (14)$$

The determinant phase $\arg \det M_q$ participates in strong CP. Under anomalous axial rephasings, phase is transported between the displayed θ -term and the mass determinant. For a common axial rephasing,

$$\theta \mapsto \theta + \Delta, \quad \arg \det M_q \mapsto \arg \det M_q - \Delta, \quad (15)$$

with sign and normalization depending on convention. Therefore

$$\bar{\theta} = [\theta + \arg \det M_q]_{2\pi} \quad (16)$$

remains invariant. Some conventions write $\theta - \arg \det(Y_u Y_d)$; the sign convention is not standing-bearing for the theorem.

Theorem 3.1 (Strong-CP quotient-survivor theorem). *In the fixed QCD/SM strong-sector scope with massive quarks, anomalous chiral transport, and nontrivial QCD topological sectors, the unique same-scope CP-odd strong-sector quotient survivor is $\bar{\theta}$. Neither θ alone nor $\arg \det M_q$ alone is the physical strong-CP endpoint object.*

Proof. Anomalous chiral rotations shift the displayed θ -coefficient and determinant phase in opposite directions. Mass-basis changes and phase conventions alter representatives but preserve the determinant quotient. The standing object after the admitted rephasing quotient is Eq. (16). This is the strong-CP lemma recorded in the Standard-Model structure corpus and is the standard field-theory invariant used in the literature. $\square$

3.3 Weak CP firewall

The CKM phase is a weak-sector mixing phase. It has its own standing in the weak/flavor ledger, and it can produce observed CP violation. It is not a strong-sector topological boundary selector. A weak CP phase cannot select $\bar{\theta}$ unless a certified determinant/descent bridge or protective mechanism is supplied. Radiative communication from weak CP into strong CP, when claimed, is therefore not primitive selection; it is a separate map requiring determinant, threshold, and stability certification.

3.4 Empirical interface

A nonzero $\bar{\theta}$ would contribute to CP-odd hadronic observables, including the neutron EDM. The current neutron EDM limit gives $|d_n| < 1.8 \times 10^{-26} e \text{ cm}$ at 90 percent confidence level [6]. Standard estimates relate this to an extremely small $|\bar{\theta}|$, roughly at the 10^{-10} level in conventional presentations [11, 15–17]. In AASC status language, this is an empirical upper constraint. It is not an internal endpoint-selection theorem.

4 From theta to CP-odd boundary load

4.1 Topological-sector bridge

Theorem 4.1 (Topological boundary-load bridge). *Let D be a fixed admissible gauge scope with nontrivial topological sectors. If a reparametrization-invariant class $\kappa \in \mathbb{R}/2\pi\mathbb{Z}$ weights those sectors and reverses under CP, then $\kappa \neq 0$ is CP-odd topological boundary load in D . Applied to QCD/SM, $\kappa = \bar{\theta}$. Hence*

$$\bar{\theta} \neq 0 \Rightarrow \text{CPoddBoundaryLoad}_{D_{\text{SCP},0}}(\bar{\theta}) = 1. \quad (17)$$

Proof. A topological-sector weighting class is not removed by local total-derivative form because the integral in Eq. (13) distinguishes global sectors. The invariant object after chiral quotienting is $\bar{\theta}$, not bare θ . Under CP the sector weight is conjugated by $\bar{\theta} \mapsto -\bar{\theta}$ modulo 2π . A nonzero class therefore distinguishes CP-conjugate topological-sector weighting, except for special sign-fixed non-identity cases handled separately below. This distinction is boundary/topological rather than local-coupling

skin, and it survives admissible redescription. Therefore nonzero $\bar{\theta}$ is CP-odd topological boundary load. $\square$

4.2 Availability is not occupancy

Proposition 4.2 (Theta-slot availability is not endpoint occupancy). *Avail_{D_{SCP,0}}($\bar{\theta} = \epsilon$) = 1 does not imply PrimitiveStanding_{D_{SCP,0}}($\bar{\theta} = \epsilon$) = 1. A θ -coordinate slot can be formally writable while a nonzero value in that slot remains unselected.*

Proof. Formal availability follows from ordinary QCD syntax and topological sectors. Primitive standing requires satisfaction of the AASC boundary and selector conditions. A coefficient may define a parameterized member of a formal family without being an internally selected endpoint of the fixed target. This is exactly the availability/occupancy distinction: writability is a comparison input, not a selection theorem. $\square$

4.3 Parameter-status taxonomy

The theorem does not ban free parameters in general. It targets nonzero CP-odd topological boundary load after same-scope quotienting.

Table 4: Parameter status: why $\bar{\theta}$ is not treated like an ordinary free parameter.

Parameter class	Example	Standing route	Why different from nonzero $\bar{\theta}$
Local interaction strength	g_s, g, g'	local transport and interaction ledger	not CP-odd topological boundary load
Mass-response carrier	Yukawa eigenvalues	matter/bridge response ledger	calibrated local-sector parameter, not sector-weighting phase
Weak CP phase	CKM phase	weak mixing holonomy and empirical weak CP witnesses	not a strong-sector topological selector
Generic local CP-violating parameter	BSM CP phase or complex Yukawa representative	local interaction or declared mixing ledger	not pure QCD topological boundary weighting unless it descends into $\bar{\theta}$
Vacuum/source parameter	cosmological constant	metric/source ledger	different source and boundary-status problem
Topological-sector phase	$\bar{\theta}$	CP-odd QCD topological boundary load	nonzero value requires selector, witness, declared input, or descent

Proposition 4.3 (Parameter non-equivalence). *The CP-odd boundary non-promotion theorem applies to nonzero topological-sector boundary load, not to arbitrary free parameters. A local coupling, mass response, or weak mixing coordinate may have standing by its own local carrier or empirical ledger. Nonzero $\bar{\theta}$ is different because its standing would introduce CP-odd topological-sector boundary load after all same-scope rephasings have been quotiented.*

5 Same-scope transport and endpoint-claim audit

5.1 Audit principle and normal forms

Every proposed route must first be typed. Same-scope transport routes act on representatives or ledgers already inside $D_{\text{SCP},0}$. Endpoint-claim routes insert a boundary value, supply empirical data, add a selector, or change target. The two classes are audited separately.

Theorem 5.1 (Same-scope strong-CP transport normal form). *Any proposed same-scope transport route affecting the strong-CP endpoint object $\bar{\theta}$ must act on at least one of the following ledgers:*

- (i) *the chiral field-basis ledger;*
- (ii) *the anomaly, Jacobian, or measure ledger;*
- (iii) *the quark-mass determinant ledger;*
- (iv) *the weak CP and flavor ledger;*
- (v) *the vacuum, vacuum-energy-response, or topological-sector ledger;*
- (vi) *the holonomy, residue, Wilson-loop, or notation ledger;*
- (vii) *the scale, scheme, or renormalization transport ledger.*

A route acting outside these transport ledgers changes the carrier inventory, boundary input, witness family, symmetry content, or scope, and is therefore not a same-scope transport route in $D_{\text{SCP},0}$.

Proof. The quotient object $\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}$ has the displayed θ -representative, the quark determinant representative, the anomalous transport law between them, the mass and weak-flavor ledgers through which representatives may be displayed, and the gauge-vacuum/topological sector on which the quotient acts. CKM and other weak-sector phases can enter only through the separate flavor ledger and require a certified typed bridge before they affect $\bar{\theta}$. A proposed operation not acting on one of these transport ledgers either adds a new carrier or symmetry, declares a boundary value, adds a positive fixing witness, or changes the comparison target. Those are $D_{\text{SCP},+}$ updates or selector imports, not same-scope transport moves. $\square$

Theorem 5.2 (Strong-CP endpoint-claim normal form). *Any proposed endpoint claim about a non-identity candidate class C_κ that is not same-scope transport must have one of the following statuses:*

- (i) *declared boundary input;*

- (ii) positive empirical fixing witness or empirical upper-bound constraint;
- (iii) richer-scope selector, protector, or descent package;
- (iv) target change, including determinant-rank collapse, added carrier, or changed witness family.

None of these is a same-scope transport operation in $D_{\text{SCP},0}$. A successful instance supplies a declared input, a witness record, or a $D_{\text{SCP},+}$ update; an unsuccessful instance is calibration import, hidden selector import, or presentation skin.

Proof. An endpoint claim that does not act on the transport ledgers of Theorem 5.1 must obtain authority from outside same-scope transport. The only available sources are boundary declaration, positive empirical fixing record, selector/protector/descent package, or target change. These classes exhaust the ways a non-identity value can be made more than a representative within the fixed quotient. Therefore they are endpoint-claim statuses, not same-scope transport moves. An unspecified “unknown mechanism” is not an endpoint-claim route: if no carrier, symmetry, boundary condition, dynamical law, witness, or descent map is supplied, it is placeholder language; if such a package is supplied, the route is a $D_{\text{SCP},+}$ update rather than a same-scope move in $D_{\text{SCP},0}$. $\square$

5.2 Mini-lemmas

Lemma 5.3 (Chiral rotation preservation). *An anomalous chiral rephasing transports phase between θ and $\arg \det M_q$ while preserving $\bar{\theta}$.*

Proof. Equation (15) gives equal and opposite shifts of the two representatives. Their quotient sum is unchanged. $\square$

Lemma 5.4 (Anomaly/Jacobian non-selector). *The anomaly and Fujikawa/Jacobian measure shift account for phase transport. They do not select a numerical value of $\bar{\theta}$.*

Proof. The anomaly equation fixes how a transformation changes the measure and representative action. It is a covariance rule. It contains no standing-bearing value-selection carrier for the invariant survivor. The anomaly-ledger sources classify such equations as bookkeeping or obstruction ledgers, not endpoint oracles [28–30]. $\square$

Lemma 5.5 (Determinant branch non-selector). *A branch choice for $\arg \det M_q$ is not a strong-CP selector unless the branch itself has certified standing.*

Proof. Changing determinant representative changes presentation of the same quotient class. A branch chosen to obtain a desired value is hidden selector import unless independently certified by mass-matrix, symmetry, or boundary data. $\square$

Lemma 5.6 (CKM firewall). *The weak-sector CKM phase does not select the strong-sector topological boundary class $\bar{\theta}$.*

Proof. CKM standing belongs to the weak mixing ledger. It may coexist with $\bar{\theta} = 0$ or $\bar{\theta} \neq 0$. Standing-bearing CP content in one ledger does not automatically grant selector authority in another

ledger. Without a typed determinant-protection, threshold, matching, or UV CP/parity bridge, it supplies no selection law for $\theta + \arg \det M_q$. $\square$

Lemma 5.7 (Weak-CP radiative non-selector). *Weak-sector CP standing can become relevant to strong CP only through a certified determinant/descent map that selects, protects, or transports a value into $\bar{\theta}$. Absent that map, CKM CP is a separate standing ledger and cannot serve as a primitive strong-sector topological boundary selector.*

Proof. Source-continuity is typed: a weak-sector CP source does not source a strong-sector topological boundary value unless a typed transport/descent map connects the ledgers. A loop, threshold, or radiative correction is an admissible transport claim only after its carrier, determinant dependence, matching map, and stability conditions are declared. Without those certificates, weak CP supplies no source-continuous selection law for nonzero strong-sector topological load. With those certificates, the route has become a selector package or richer-scope descent and is no longer a same-scope primitive endpoint in $D_{\text{SCP},0}$. $\square$

Lemma 5.8 (Vacuum relabeling non-erasure). *θ -vacuum or residual-sector relabeling does not erase the quotient survivor $\bar{\theta}$.*

Proof. A vacuum representative may change the label used for sector decomposition, but the topological-sector weighting class remains the object read by CP-odd strong-sector observables. Gauge-vacuum and residual-sector sources block representative change from becoming boundary selection without a certified residual selector [33–35]. $\square$

Lemma 5.9 (Vacuum-energy profile non-selector). *The QCD vacuum-energy dependence $E(\bar{\theta})$, when defined, is a response function over parameterized topological sectors. It is not a same-scope selection law for $\bar{\theta}$ in fixed QCD/SM, because $\bar{\theta}$ is not a dynamical coordinate in $D_{\text{SCP},0}$. Minimization of $E(\bar{\theta})$ becomes a selector only after a dynamical relaxation carrier is added, as in PQ/axion scope.*

Proof. A response profile can say how the theory behaves for each boundary class. It does not move the boundary class unless that class is promoted to a field, constrained by a standing boundary law, or selected by an additional carrier. In ordinary fixed QCD/SM, $\bar{\theta}$ labels the sector weighting; it is not varied by an equation of motion. Therefore the vacuum-energy profile is not a primitive same-scope endpoint selector in $D_{\text{SCP},0}$. An axion supplies precisely the missing dynamical coordinate and is consequently a richer-scope selector package rather than a same-scope vacuum-energy argument [5, 7–10]. $\square$

Lemma 5.10 (Holonomy notation non-selector). *Wilson-loop, Chern-Simons, residue, or holonomy notation does not select $\bar{\theta}$ unless the residue class is standing-certified.*

Proof. Holonomy notation can expose a topological residue, but notation itself is not a selector. The discrete holonomy and curvature-obstruction sources require quotient-stable, boundary-detectable, transport-witnessed residue before promotion [26, 31, 32]. $\square$

Lemma 5.11 (Renormalization non-fixation). *Scale or scheme transport does not convert $\bar{\theta}$ into a fixed endpoint value.*

Proof. Renormalization changes descriptions across scale or scheme. It does not by itself supply CP-odd boundary selection. A value fixed by hand at one scale remains declared input unless protected or dynamically selected. $\square$

Lemma 5.12 (EDM upper-bound non-derivation). *The neutron EDM upper bound is an empirical constraint $\text{CPBound}(\bar{\theta})$. It is not a positive fixing witness $\text{WitFix}(C_\kappa)$ and does not derive $\bar{\theta} = 0$.*

Proof. An upper bound restricts the set of phenomenologically viable reconstructions. It neither identifies a nonzero class C_κ nor certifies the identity endpoint. Promoting it to an exact theoretical endpoint would be calibration import, forbidden by the exactness and calibration-removal sources [24]. $\square$

Lemma 5.13 (Richer-scope recognition). *PQ/axion, UV CP/parity, Nelson-Barr, flavor/modular, determinant-rank/massless-quark, and positive-EDM routes are not same-scope erasures. They add a selector, alter the target, collapse the determinant-rank ledger, or supply witness data.*

Proof. Each route modifies the carrier inventory, boundary input, field content, symmetry ledger, positive-witness/update ledger, determinant-rank status, or mass spectrum. An exact massless-quark route makes the determinant phase ill-defined or renders $\bar{\theta}$ unphysical by changing the massive-quark target; it is not a same-scope selector for $D_{\text{SCP},0}$. Such additions are admissible only as declared scope changes, determinant-rank target changes, or witness updates. $\square$

Theorem 5.14 (Same-scope strong-CP transport exhaustion). *Inside $D_{\text{SCP},0}$, every alleged same-scope transport route for setting, erasing, or explaining nonzero $\bar{\theta}$ is invariant-preserving transport, hidden selector, presentation skin, or target-changing import. Endpoint-claim routes are separately typed as declared input, empirical constraint or fixing witness, richer-scope selector/descent, or target update. No same-scope transport route satisfies $\text{SelectorPkg}_\theta^{\text{cert}}$.*

Proof. The transport normal-form theorem reduces any same-scope proposal to the chiral, anomaly, determinant, weak/flavor, vacuum, vacuum-energy-response, topological, holonomy/residue, or scale/scheme ledgers. Lemmas 5.3–5.11 audit those ledgers: chiral, anomaly, determinant, CKM, radiative-weak, vacuum-relabeling, vacuum-energy-profile, holonomy, and renormalization moves either preserve the invariant or fail to supply certified selection. The endpoint-claim normal-form theorem then types EDM bounds, declared values, richer-scope mechanisms, determinant-rank changes, and alleged unknown stabilizers as constraints, inputs, witnesses, selector packages, target updates, or placeholder language rather than same-scope transport. Lemmas 5.12–5.13 audit those endpoint-claim statuses. Any successful mechanism adds carrier load, boundary input, witness data, determinant-rank collapse, or symmetry/descent structure, and is therefore a $D_{\text{SCP},+}$ update rather than a same-scope operation. An unspecified stabilizer performs no source-continuity work and does not reopen the fixed target. $\square$

Appendix A records the corresponding failure-route ledger in operational form. Each route is classified there as invariant-preserving transport, endpoint-claim status, declared input, richer-scope selector, target update, hidden selector, presentation skin, or inert bookkeeping.

6 Selector burden and CP-odd boundary no-promotion

6.1 Boundary source continuity

Lemma 6.1 (Non-neutral boundary distinction). *Let b be a boundary class in a fixed target D . If b distinguishes admissible sector weights, endpoint histories, orientation classes, or comparison outcomes, then b carries non-neutral boundary load:*

$$b \neq 0 \wedge \text{sector-weighting distinction} \Rightarrow \text{NonNeutralBoundaryLoad}_D(b) = 1. \quad (18)$$

Proof. A boundary class is neutral only when replacing it by the identity leaves the sector weighting and endpoint comparison ledger unchanged. If a class changes the relative weight of sectors, distinguishes oriented histories, or changes endpoint comparison content, then it performs work in the boundary ledger. Such work is non-neutral load rather than presentation skin. $\square$

Lemma 6.2 (Unsourced non-neutral boundary load). *If non-neutral boundary load in a fixed admissible domain has no selector, declared input, positive empirical fixing witness, richer-scope descent, or selecting law, then it cannot have primitive standing in that domain.*

Proof. Primitive standing requires reference, standing, admissibility, and irreversibility to be preserved as fixed-domain roles. Unsupported non-neutral boundary load fails each role in turn: reference has no source by which the load belongs to the same target; standing cannot license reuse of a distinction with no carrier; admissibility cannot separate the distinction from branch choice; and irreversibility cannot prevent later repair by hidden selector or post-hoc input. If the load is instead stipulated as primitive reference data, then it has been converted into declared boundary input and exits the primitive fixed-scope endpoint target. Restoring any lost role imports exactly the missing source work. Therefore unsourced non-neutral boundary load is not primitive standing. $\square$

Lemma 6.3 (CP-odd topological load is non-neutral). *A nonzero CP-odd topological-sector weighting class is non-neutral boundary load. In particular, generic nonzero $\bar{\theta}$ changes CP-conjugate sector weighting in the strong-CP target.*

Proof. Under CP, $Q_{\text{top}} \mapsto -Q_{\text{top}}$ and a generic phase weight $e^{i\kappa Q_{\text{top}}}$ is replaced by $e^{-i\kappa Q_{\text{top}}}$. If κ is not the identity class, the sector weighting is changed. Thus a generic nonzero CP-odd κ is not neutral notation but boundary load. Taking $\kappa = \bar{\theta}$ gives the strong-CP specialization after the chiral quotient. $\square$

Theorem 6.4 (CP-odd boundary source continuity). *In a fixed admissible domain D , any standing-bearing nonzero CP-odd boundary distinction must be continuous with at least one upstream admissible source: a certified selector package, declared boundary input, positive empirical CP-odd fixing witness, richer-scope descent, or symmetry/dynamical law selecting the distinction.*

Proof. By Lemma 6.3, a nonzero CP-odd topological boundary distinction is non-neutral boundary load. By Lemma 6.2, such load cannot have primitive standing without source continuity. The source classes are exactly selector package, declared boundary input, positive empirical fixing witness, richer-scope descent, or an included selecting law. $\square$

6.2 Non-identity topological load

Theorem 6.5 (Non-identity topological boundary-load burden). *For the freely variable strong-CP quotient object under audit, every non-identity topological-sector weighting class requires selector, declared input, positive empirical fixing witness, or richer-scope descent when promoted as primitive endpoint standing. Generic nonzero classes carry CP-odd load. The class π is sign-fixed under CP modulo 2π , but it remains non-identity topological load. Thus, for the candidate classes of Definition 0.5,*

$$\begin{aligned} \forall \kappa \in \mathbb{R}/2\pi\mathbb{Z}, \quad \kappa \neq 0 \wedge \text{FreeTopPhase}_{D_{\text{SCP},0}}(C_\kappa) = 1 \\ \Rightarrow \text{NonIdentityTopologicalLoad}_{D_{\text{SCP},0}}(C_\kappa) = 1. \end{aligned} \quad (19)$$

In $D_{\text{SCP},0}$, every nonzero C_κ is freely variable unless a source status is supplied. With no selector, input, positive fixing witness, or descent, every nonzero C_κ is deleted from the certified primitive survivor set.

Proof. For any $\kappa \in \mathbb{R}/2\pi\mathbb{Z}$, the topological-sector weight is $e^{i\kappa Q_{\text{top}}}$. The identity class 0 gives weight one in every sector. Every other freely variable class changes the sector-weighting ledger without being fixed by a consistency theorem, quantization law, source, or selector. Generic nonzero classes are exchanged with CP-conjugate classes and are CP-odd boundary load; π is special only because it is sign-fixed, not because it is the identity. Since non-identity topological load is non-neutral boundary load, Lemma 6.2 applies. In $D_{\text{SCP},0}$, the admissible source classes are absent by target definition and transport exhaustion, so no freely variable non-identity candidate class survives as primitive endpoint standing. $\square$

Remark 6.6 (Scope of the non-identity burden). *This theorem is not a general prohibition on all nonzero topological terms. It applies to primitive promotion of a freely variable non-identity topological-sector phase, in the sense of Definition 0.7, in the strong-CP quotient object $\bar{\theta} \in \mathbb{R}/2\pi\mathbb{Z}$. Quantized topological coefficients, anomaly-fixed terms, Wess-Zumino terms, Chern-Simons levels, and discrete topological data fixed by gauge consistency, global structure, anomaly matching, or declared boundary law occupy a different AASC status: their source is the corresponding consistency, carrier, or boundary theorem. The present result concerns nonzero primitive strong-CP boundary standing in $D_{\text{SCP},0}$.*

Lemma 6.7 (Uniform non-identity deletion). *The deletion of nonzero endpoint classes is uniform over the continuum of candidates:*

$$\begin{aligned} \forall \kappa \neq 0, \quad \text{NonIdentityTopologicalLoad}_{D_{\text{SCP},0}}(C_\kappa) \wedge \neg \text{SelectorPkg}_{\bar{\theta}}^{\text{cert}} \wedge \neg \text{Input}_{D_{\text{SCP},0}}(C_\kappa) \\ \wedge \neg \text{WitFix}_{D_{\text{SCP},0}}(C_\kappa) \wedge \neg \text{Rich}_{D_{\text{SCP},0}}(C_\kappa) \Rightarrow \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0. \end{aligned} \quad (20)$$

The proof does not enumerate values of κ ; it applies the same non-identity-load and source-burden predicate to every nonzero class.

Proof. The antecedent contains exactly the facts used by Theorem 6.5: non-identity topological load and absence of certified source. These facts are independent of the arithmetic representation of κ . Therefore the deletion is schematic and uniform across the family $\{C_\kappa : \kappa \neq 0\}$. $\square$

6.3 Selector-burden theorem

Theorem 6.8 (Nonzero- $\bar{\theta}$ selector-burden theorem). *For each nonzero candidate class C_κ in the fixed target $D_{\text{SCP},0}$, primitive endpoint standing requires*

$$\begin{aligned} \kappa \neq 0 \wedge \text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_\kappa) \Rightarrow & \text{SelectorPkg}_{\bar{\theta}}^{\text{cert}} \vee \text{Input}_{D_{\text{SCP},0}}(C_\kappa) \\ & \vee \text{WitFix}_{D_{\text{SCP},0}}(C_\kappa) \vee \text{Rich}_{D_{\text{SCP},0}}(C_\kappa). \end{aligned} \quad (21)$$

A written θ -term, chiral-basis choice, determinant branch, anomaly equation, regulator convention, θ -vacuum label, vacuum-energy profile, or EDM upper bound does not satisfy this burden.

In ordinary physics language, this is the “new physics” burden: a nonzero $\bar{\theta}$ endpoint requires something that carries its selection or stabilization. The phrase “new physics” is used here as a gloss for a certified source-continuity package, not as a separate premise. Positive empirical fixing witnesses are part of the formal reopening condition because they update the target; they are evidence or records rather than dynamical mechanisms in the narrow BSM sense.

Proof. By Theorem 6.5, every C_κ with $\kappa \neq 0$ is non-identity topological boundary load; generic nonzero classes are CP-odd load, while C_π remains non-identity load. By Theorem 6.4, standing-bearing CP-odd boundary distinction requires a typed source, and by the non-identity theorem the sign-fixed π class requires the same source because it is not the identity. The allowed source statuses are exactly selector package, declared input, positive empirical fixing witness, richer-scope descent, or selecting law included in the package. The listed non-examples are representative, branch, transport, notation, response-profile, or constraint artifacts; the same-scope audit shows that none certifies the package. $\square$

6.4 Primitive CP-odd boundary load impossibility

Theorem 6.9 (Primitive non-identity boundary load impossibility). *In $D_{\text{SCP},0}$, if*

$$\neg \text{SelectorPkg}_{\bar{\theta}}^{\text{cert}} \wedge \forall \kappa \neq 0 \neg \text{Input}_{D_{\text{SCP},0}}(C_\kappa) \wedge \forall \kappa \neq 0 \neg \text{WitFix}_{D_{\text{SCP},0}}(C_\kappa) \wedge \forall \kappa \neq 0 \neg \text{Rich}_{D_{\text{SCP},0}}(C_\kappa), \quad (22)$$

then

$$\forall \kappa \in \mathbb{R}/2\pi\mathbb{Z}, \kappa \neq 0, \quad \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0. \quad (23)$$

Proof. Assume some C_κ with $\kappa \neq 0$ survives as primitive fixed-scope standing in $D_{\text{SCP},0}$. Theorem 6.8 then requires selector package, declared input, positive fixing witness, or richer descent for that class. These are all negated by Eq. (22). Lemma 6.7 supplies the uniform deletion over the full continuum of non-identity candidates. Therefore no non-identity C_κ is a certified primitive same-target survivor. $\square$

6.5 Identity-class certification

Proposition 6.10 (Identity-class availability). *The identity candidate class is available in the strong-CP quotient:*

$$\text{Avail}_{D_{\text{SCP},0}}(C_0) = 1. \quad (24)$$

Proof. The quotient-survivor theorem places $\bar{\theta}$ in $\mathbb{R}/2\pi\mathbb{Z}$. Every quotient group contains its identity element, and here that identity is 0 modulo 2π . Therefore C_0 is available in the same object space as every proposed nonzero endpoint class. $\square$

Proposition 6.11 (Identity-class primitive standing). *The identity candidate class has primitive standing in $D_{\text{SCP},0}$:*

$$\text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_0) = 1. \quad (25)$$

Proof. The candidate class C_0 is not an additional CP-odd boundary datum. It is the absence of non-identity topological-sector weighting in the admitted quotient, expressed by $e^{i0Q_{\text{top}}} = 1$ for every sector. Its standing is inherited from the admitted quotient object $\bar{\theta} \in \mathbb{R}/2\pi\mathbb{Z}$, not from a separately stipulated zero coefficient. It requires no CP-odd or non-identity selector because it carries no CP-odd or non-identity boundary load. It is therefore primitive in the fixed target rather than supplied by a hidden branch choice. $\square$

Proposition 6.12 (Identity endpoint is not a coefficient-zero branch). *The selected endpoint C_0 is not obtained by choosing $\theta = 0$, choosing $\arg \det M_q = 0$, or stipulating a displayed coefficient zero. It is obtained only after quotienting to θ , deleting every non-identity candidate C_κ by the non-identity burden theorem, and certifying the quotient identity as available primitive standing. Hence*

$$\text{ChosenCoeffZero}(\theta) \not\equiv \text{Sel}_{D_{\text{SCP},0}}(C_0). \quad (26)$$

Proof. A coefficient-zero branch acts on a representative. The endpoint theorem acts on the quotient family C_κ . Representative zeros can be moved by chiral transport and therefore have no endpoint authority by themselves. The identity endpoint is selected only after all non-identity quotient classes fail the certified-survivor test and the identity class satisfies availability and primitive-standing conditions. $\square$

Theorem 6.13 (Identity-class non-selection). *The zero candidate class C_0 is not a selected nonzero boundary value. It is the identity class carrying no non-identity topological boundary load. If every non-identity topological boundary class is non-promoted in D , the unique primitive endpoint class in that target is C_0 .*

Proof. The identity element of $\mathbb{R}/2\pi\mathbb{Z}$ contributes no phase factor distinguishing topological sectors: $e^{i0Q_{\text{top}}} = 1$. Propositions 6.10 and 6.11 certify the two positive conditions required for identity-class selection, and Proposition 6.12 separates that selection from a representative coefficient-zero choice. Theorem 6.5 deletes every non-identity class when no source exists. The remaining endpoint is the identity class. $\square$

6.6 Fixed-scope strong-CP endpoint

Theorem 6.14 (Strong-CP fixed-scope endpoint theorem). *In the fixed strong-CP target $D_{\text{SCP},0}$,*

$$\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1, \quad \forall \kappa \in \mathbb{R}/2\pi\mathbb{Z}, \kappa \neq 0, \quad \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0. \quad (27)$$

Equivalently, in standard strong-CP notation, the selected fixed-scope endpoint is $\bar{\theta} = 0$.

Proof. The quotient-survivor theorem identifies $\bar{\theta}$ as the object of audit and Definition 0.5 decomposes its endpoint candidates into the family $\{C_\kappa\}_{\kappa \in \mathbb{R}/2\pi\mathbb{Z}}$. The non-identity load theorem shows that every C_κ with $\kappa \neq 0$, including C_π , is non-identity topological boundary load. The transport and endpoint-claim exhaustion theorems show that $D_{\text{SCP},0}$ contains no same-scope transport or endpoint-claim route satisfying the selector package. CKM CP is already included in $D_{\text{SCP},0}$ as a separate standing ledger; by Lemma 5.7, typed source-continuity does not let weak-sector CP source a strong topological boundary endpoint without a certified determinant/descent map. By definition of $D_{\text{SCP},0}$, there is no declared nonzero boundary input, no positive empirical strong-sector fixing witness, and no richer-scope descent. The primitive-load theorem deletes every C_κ with $\kappa \neq 0$ from the certified primitive survivor set. Proposition 6.10 proves $\text{Avail}_{D_{\text{SCP},0}}(C_0) = 1$, and Proposition 6.11 proves $\text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_0) = 1$. Criterion 2.10 therefore gives $\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1$, i.e. $\bar{\theta} = 0$ in ordinary notation. $\square$

7 Declared input, $\theta = \pi$, and parameterized QCD

7.1 Declared-input compatibility

Theorem 7.1 (Declared-input compatibility). *The formal family $\text{QCD}_{\bar{\theta}=\epsilon}$ is definable for $\epsilon \in \mathbb{R}/2\pi\mathbb{Z}$. If $\epsilon \neq 0$, then ϵ has one of the following AASC statuses: declared external boundary input, positive-witness target, richer-scope descent, or selected/protected value. It is not an internally selected primitive endpoint of $D_{\text{SCP},0}$, and such a parameterized target is not a same-target counterexample to $D_{\text{SCP},0}$.*

Proof. Formal definability is availability. Primitive fixed-scope selection is stronger. A declared $\epsilon \neq 0$ changes the target statement by inserting a boundary datum; it can be studied as such. The resulting parameterized target is a different status object or update, not a same-target counterexample to the primitive endpoint theorem. Without that declaration or another source, Theorem 6.14 applies to $D_{\text{SCP},0}$. $\square$

7.2 The $\theta = \pi$ class

Proposition 7.2 (The $\theta = \pi$ non-identity caveat). *The class $\pi \in \mathbb{R}/2\pi\mathbb{Z}$ is sign-fixed under $\theta \mapsto -\theta$ modulo 2π , but it is not the identity class. It changes topological-sector weighting and may carry nontrivial vacuum-sector structure. Therefore it is not admitted by load absence and is deleted by the non-identity load theorem unless selected, declared, witnessed, or inherited. If a theory selects $\bar{\theta} = \pi$, the selecting vacuum-sector or symmetry structure must be declared.*

Proof. Although $\pi = -\pi$ modulo 2π , $e^{i\pi Q_{\text{top}}} = (-1)^{Q_{\text{top}}}$, not the identity sector weight. Thus π is not load-free. Its CP-sign-fixed status does not turn it into the zero endpoint. Theorem 6.5 therefore applies: an admissible π -endpoint requires a selector, witness, or declared boundary rule specifying that non-identity topological structure. $\square$

8 Richer-scope selector normal form

The phrase “new physics” is useful here but should be read structurally. In the narrow model-building idiom it usually denotes new fields, symmetries, boundary conditions, determinant-protection laws, or dynamical relaxation coordinates. In the formal reopening condition used here, the same burden is stated more generally: some certified source-continuity package must carry the nonzero or dynamically relaxed strong-CP class, whether by new dynamical structure, declared boundary input, positive empirical fixing witness, or matching/descent map. Positive fixing witnesses are evidence that force target reconstruction; they are not themselves dynamical mechanisms. The theorem does not rule out such structures. It says they must be supplied and audited; they cannot be invoked as an unspecified stabilizer while remaining inside $D_{\text{SCP},0}$.

8.1 General certificate

Any successful non-fixed-scope solution must provide

$$\mathcal{C}_{\text{SCP}} = (\Sigma_{\bar{\theta}}, D \hookrightarrow D', \text{Transport}_{D'}, \text{Selection}_{D'}, \text{Stability}_{D'}, \text{Descent}_{D' \rightarrow D_{\text{IR}}}). \quad (28)$$

Without this certificate, a proposed solution is reclassification, hidden selector import, calibration, or notation.

Table 5: Richer-scope selector certificate matrix.

Mechanism	Carrier	Selection mode	Stability burden	Descent burden
PQ/axion	axion field plus PQ symmetry	dynamic relaxation of $\bar{\theta}_{\text{eff}}$	axion quality, gravitational breaking, thresholds	low-energy minimum compatible with EDM limits
UV CP	CP symmetry and breaking sector	protective initial zero plus controlled determinant phase	CP-breaking radiative corrections	CKM allowed while $\bar{\theta}$ remains controlled
UV parity	parity or left-right structure	parity protection of strong CP	parity-breaking thresholds	QCD/SM descent after symmetry breaking
Nelson-Barr	vectorlike quarks/determinant structure	CKM without determinant-phase pollution	loop stability and texture protection	low-energy determinant remains controlled
Flavor/modular	flavor or modular carrier	determinant or texture constraint	deformation and symmetry-breaking stability	physical masses/mixings descend correctly
Massless quark	determinant-rank collapse / exact zero quark mass	makes $\arg \det M_q$ ill-defined or $\bar{\theta}$ unphysical by exact chiral freedom	empirical and lattice compatibility plus exact-rank certification	singular target change from the massive-quark $D_{\text{SCP},0}$ target
Future nonzero EDM	positive strong-sector CP-odd fixing record	witness fixation	experimental and systematic stability	theory reconstruction

8.2 Peccei-Quinn and axion relaxation

In a PQ/axion theory,

$$\bar{\theta}_{\text{eff}} = \theta + \arg \det M_q + \frac{a}{f_a}. \quad (29)$$

The axion is a new carrier. Its potential dynamically selects a CP-conserving minimum in standard treatments [7–12]. AASC classifies this as a richer-scope dynamic selector, not a same-scope erasure. The axion quality problem is precisely the stability burden on the selector [18].

8.3 UV CP, parity, Nelson-Barr, and flavor routes

UV CP/parity mechanisms and Nelson-Barr mechanisms can be legitimate only when the carrier, determinant protection, breaking pattern, loop stability, and low-energy descent are certified [13, 14]. Flavor or modular constructions face the same burden: a mass-matrix texture is not a selector unless the texture itself has independent standing and survives deformations.

8.4 Empirical witness update

A future nonzero EDM attributable to strong CP would not refute the fixed-target theorem. It would instantiate $D_{\text{SCP},+}$, adding a positive strong-sector CP-odd fixing witness. The task would then be reconstruction: identify whether the witness descends from declared nonzero input, a richer-scope selector, additional CP-odd operators, or another certified target update. Thus the theorem does not claim that nature cannot contain nonzero strong CP. It claims that, if nature does, the nonzero value is not primitive standing inside $D_{\text{SCP},0}$; it is evidence for a target update and a reconstruction burden.

9 Same-target closure and counterexample discipline

9.1 AASC dependence as theorem class

AASC dependence fixes the theorem class. A contrary same-target construction must invalidate a definition, theorem link, dependency transport, source certification, target specialization, or final inference; alternatively, it must derive a load-bearing kernel role from below or realize a faithful same-regime counterexample.

A demand that the result be restated as a conventional naturalness claim changes the proof class and does not contest the same theorem. The theorem is evaluated as an AASC endpoint theorem over the fixed target $D_{\text{SCP},0}$, not by converting it into a conventional naturalness claim. It classifies the standing of nonzero topological boundary load inside that target.

9.2 Cost of kernel denial

Lemma 9.1 (Cost of kernel denial). *A denial of the AASC kernel while retaining the same strong-CP endpoint target either concedes equivalent governance work under different vocabulary or abandons a target-adequacy condition. Abandoning the condition incurs the corresponding loss in Table 6.*

Table 6: Cost of kernel denial in $D_{\text{SCP},0}$.

Denied adequacy condition	Kernel role lost	Cost to the strong-CP target
Target determinacy	Reference	θ , $\arg \det M_q$, $\bar{\theta}$, EDM bounds, axion relaxation, declared parameter families, and C_κ candidates no longer remain distinct target objects.
Status evaluability	Standing	Formal availability, quotient survival, declared boundary input, primitive endpoint standing, selector descent, positive fixing witness, upper-bound/null-channel constraint, empirical update, and placeholder mechanism language collapse into one undifferentiated status.
Act-time finality	Irreversibility	A failed nonzero C_κ can be repaired after audit by importing axion/PQ, UV CP, CKM descent, declared input, or empirical witness while pretending to remain the same fixed-scope endpoint.
Same-regime fidelity	Admissibility boundary	Chiral rotation, anomaly bookkeeping, determinant branch choice, vacuum-energy response, same-scope transport, richer-scope selector import, and target update become interchangeable.

Theorem 9.2 (No cost-free kernel denial). *A same-target objection that denies the kernel either reintroduces equivalent governance work under different vocabulary or abandons the determinate strong-CP endpoint target. There is no third position from which one may reject the kernel while preserving the same fixed-scope endpoint evaluation.*

Proof. If the objection preserves target determinacy, status evaluability, act-time finality, and same-regime fidelity, then by Theorem 2.3 it has preserved the governance work of reference, standing, irreversibility, and admissibility. If it denies one of those conditions, Table 6 records the corresponding loss. A later device that restores the lost good performs governance-equivalent work or changes target. Thus kernel refusal is either terminological or target-changing. $\square$

9.3 No strict same-target weakening

Definition 9.3 (Core strong-CP endpoint adequacy packet). *The core packet $\mathcal{C}_{0,\text{SCP}}$ consists of the fixed target $D_{\text{SCP},0}$, the kernel roles Adm , Stand , Ref , Irr , the coordinate slot Θ_{slot} , the quotient object $\bar{\theta}$, the candidate family C_κ , the selector-burden criterion, the same-scope transport audit, the endpoint-claim audit, and the identity-class selection criterion.*

Definition 9.4 (Nonzero-admission-permissive weakening). *A nonzero-admission-permissive weakening $W_{\bar{\theta}}$ is a purported weaker regime preserving $\mathcal{C}_{0,\text{SCP}}$ while allowing some $C_\kappa \neq C_0$ to have*

primitive endpoint standing without selector package, declared boundary input, positive fixing witness, or richer-scope descent.

Theorem 9.5 (No strict same-target weakening permits nonzero primitive strong CP). *No strict weakening preserves $D_{\text{SCP},0}$, the quotient survivor $\bar{\theta}$, the non-identity boundary-load bridge, fixed-domain boundary standing, and transport/endpoint-claim exhaustion while certifying $\text{PrimitiveStanding}_{D_{\text{SCP},0}}(C_\kappa) = 1$ for any $\kappa \neq 0$ without selector package, declared input, positive fixing witness, or richer descent.*

Proof. Assume $W_{\bar{\theta}}$ exists. If it weakens reference, then C_κ is no longer the same candidate family. If it weakens standing, formal availability becomes endpoint selection and the theorem no longer evaluates primitive standing. If it weakens admissibility, same-scope transport and scope update collapse, so axion/PQ, UV CP, declared input, CKM descent, and vacuum-response routes can be imported without target accounting. If it weakens irreversibility, a failed nonzero candidate can be repaired by later selector import while counted as the same act. If it adds a middle status, that status either affects endpoint standing, in which case it is hidden selector work, or affects nothing, in which case it is bookkeeping. If it preserves all kernel work and all members of $\mathcal{C}_{0,\text{SCP}}$, then Theorems 6.8–6.14 still apply and it is not a strict weakening. Every case either changes target, imports governance-equivalent work, becomes inert, or preserves the endpoint theorem. $\square$

9.4 Faithful same-regime counterexample conditions

Theorem 9.6 (Faithful same-regime counterexample conditions). *A faithful same-regime counterexample must preserve $D_{\text{SCP},0}$, the quotient survivor $\bar{\theta}$, the non-identity topological boundary-load classification, the fixed boundary standing rules, and same-scope transport plus endpoint-claim exhaustion, while producing certified primitive nonzero $\bar{\theta}$ without selector package, declared input, positive empirical fixing witness, or richer-scope descent. An example that changes field content, symmetry content, boundary input, positive-witness/update ledger, or target quotient is a $D_{\text{SCP},+}$ update rather than a counterexample.*

9.5 Right-mode objection targets

A same-target objection to the endpoint theorem must attack at least one load-bearing link below. Objections outside these classes are non-standing unless they identify a theorem-level failure in one of the cited bridges.

- (R1) **Kernel-neutral target adequacy:** show that determinate strong-CP endpoint evaluation does not require target determinacy, status evaluability, act-time finality, or same-regime fidelity.
- (R2) **Kernel forcing:** show that those adequacy roles do not force reference, standing, admissibility, and irreversibility.
- (R3) **Context non-smuggling:** show that $D_{\text{SCP},0}$ secretly asserts C_0 or deletes $C_\kappa \neq C_0$ before proof.
- (R4) **QCD quotient bridge:** show that $\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}$ is not the strong-CP quotient survivor.

- (R5) **Boundary-load bridge:** show that nonzero C_κ is not freely variable non-identity topological boundary load in $D_{\text{SCP},0}$.
- (R6) **Selector-burden theorem:** show that non-identity topological boundary load can have primitive endpoint standing without selector, declared input, positive fixing witness, or richer-scope descent.
- (R7) **Same-scope transport normal form:** produce a same-scope transport route outside chiral, anomaly, determinant, CKM, vacuum/topological, holonomy/residue, or renormalization transport.
- (R8) **Endpoint-claim normal form:** produce an endpoint-claim route outside declared input, positive fixing witness, upper-bound/null-channel record, richer-scope selector, determinant-rank change, or target update.
- (R9) **Unspecified stabilizer route:** show a certified source-continuity package for $C_\kappa \neq C_0$. A bare assertion that an unknown mechanism might stabilize a small nonzero value is not a same-target objection until the carrier, law, boundary condition, witness, or descent map is specified.
- (R10) **Vacuum-energy non-selector:** show that $E(\bar{\theta})$ selects $\bar{\theta}$ inside fixed QCD/SM without a dynamical carrier.
- (R11) **CKM typed source-continuity:** show that weak-sector CP supplies strong-sector topological boundary standing without determinant/descent bridge.
- (R12) **Identity-class certification:** show that C_0 is not available, not primitive-standing, or is merely chosen coefficient-zero.
- (R13) **Family-wise deletion:** show that some nonzero C_κ survives while preserving the fixed target and without importing a selector/input/witness/descent package.
- (R14) **Selection criterion:** show that the final endpoint violates Criterion 2.10.
- (R15) **Declared-input compatibility:** show that $\text{QCD}_{\bar{\theta}=\epsilon}$ with $\epsilon \neq 0$ is the same primitive endpoint target rather than declared boundary input or target update.
- (R16) **Kernel weakening:** exhibit a strict weakening preserving $\mathcal{C}_{0,\text{SCP}}$ while allowing nonzero primitive standing.

9.6 Standing-objection register

Table 7: Standing-objection register.

Objection	Standing	Response
QCD allows theta.	Partial	Granted. Writability is availability, not primitive endpoint standing.
$\bar{\theta}$ is physical.	Granted	Exactly. The theorem audits the physical quotient survivor, not a removable representative.

Objection	Standing	Response
This defines zero into the result.	Serious if no burden theorem	The selector-burden and source-continuity theorems derive non-promotion for non-identity boundary load; C_0 is certified by availability and primitive-standing propositions.
Free parameters are common.	Serious if overbroad	The theorem is narrow: nonzero CP-odd topological boundary load after quotienting same-scope rephasings.
Nonzero-theta QCD is valid.	Granted as availability	It is a declared parameterized family member or boundary insertion, not the primitive endpoint of $D_{\text{SCP},0}$.
What about $\theta = \pi$?	Technical	π is sign-fixed but non-identity topological load; it requires selector or declared vacuum structure.
Axions solve strong CP.	Compatible	Axions are richer-scope dynamic selectors, exactly as the theorem predicts.
This is only naturalness.	No	No probability, typicality, or fine-tuning prior appears. The theorem is a standing/exhaustion result.
AASC is unconventional.	Not a same-target objection	AASC dependence fixes theorem class; a counterexample must break a load-bearing dependency or preserve the target while reversing the endpoint.
The theorem is tautological once AASC is accepted.	No, unless a bridge theorem is collapsed into a definition	The kernel fixes the proof calculus. The endpoint still requires quotient identification, candidate-family formation, non-identity load classification, route exhaustion, selector-burden failure, and identity certification.
No new QCD dynamics or predictions are supplied.	Correct but non-standing	The theorem class is endpoint standing and no-promotion, not model-building, lattice calculation, or phenomenological forecasting.
Maybe an unknown mechanism stabilizes a small nonzero $\bar{\theta}$.	Not yet	A reopening condition requires a specified source-continuity package: carrier, law, symmetry, boundary condition, positive fixing witness, or descent map. Without that package, the claim is placeholder language. With it, the target updates to $D_{\text{SCP},+}$.

10 Conclusion

The strong-CP problem is not solved by finding a better chiral basis. The invariant $\bar{\theta}$ is precisely what survives chiral transport, anomaly bookkeeping, determinant rephasing, weak-sector rephasing, vacuum relabeling, holonomy notation, and scale transport. The AASC question is sharper: what

gives a nonzero CP-odd QCD topological boundary class standing?

The answer is not formal writability. The QCD-to-boundary bridge shows that generic nonzero $\bar{\theta}$ is CP-odd topological boundary load, and the non-identity burden theorem shows that every C_κ with $\kappa \neq 0$ is non-identity topological load. The selector-burden theorem shows that such load cannot be primitive in the fixed strong-CP target without a certified selector package, declared input, positive empirical CP-odd fixing witness, or richer-scope descent. Transport and endpoint-claim exhaustion find no such package in $D_{\text{SCP},0}$. The identity-class theorem shows that C_0 is not a secretly chosen CP-odd value; it is the unique load-free identity endpoint. Therefore

$$\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1, \quad \forall \kappa \neq 0, \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0.$$

Axions, UV CP/parity, Nelson-Barr constructions, flavor/modular determinant routes, declared nonzero boundary inputs, determinant-rank/massless-quark target changes, and future positive EDM fixing records are not refutations of the theorem. They are precisely the selector, input, positive-witness, or target-update packages the theorem says are required. In the usual physics idiom, the result can be stated compactly: without new physics, the fixed strong-CP target selects $\bar{\theta} = 0$. More exactly, absent a certified selector, declared nonzero boundary input, positive empirical fixing witness, or richer-scope source-continuity package, no $C_\kappa \neq C_0$ carries primitive endpoint standing. Positive fixing witnesses are included in that formal reopening condition as target-updating evidence, not as dynamical new physics in the narrow model-building sense. The theorem does not prove that nonzero $\bar{\theta}$ is impossible in nature. It proves that a nonzero value requires specified source continuity. Vague appeal to an unknown stabilizer is not a mechanism; once the stabilizer is specified, it is a target update, not a same-target counterexample.

The endpoint is AASC-internal in the precise necessity sense: inside any non-degenerate regime of claim-bearing endpoint use for the fixed strong-CP target, the kernel roles of reference, standing, admissibility, and irreversibility are already active, and the AASC consequence layer audits those forced roles rather than adding an optional external classifier. That dependency is explicit rather than concessive. A contrary same-target construction must derive a kernel role from below, break the strong-CP specialization of a load-bearing kernel dependency, identify a source-audit failure, supply a certified selector package inside $D_{\text{SCP},0}$, or satisfy the faithful same-target counterexample conditions. Without one of those results, refusing the vocabulary of AASC either concedes the same governance work under another name or abandons the fixed strong-CP endpoint target. The endpoint remains $\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1$.

A Failure-route table

Route	Standard move	AASC classification	Result
Chiral rotation	remove displayed theta from one term	invariant-preserving transport	$\bar{\theta}$ survives
Anomaly compensation	use measure shift	bookkeeping	no value selection
Mass diagonalization	choose mass basis	redescription skin	determinant class preserved

Route	Standard move	AASC classification	Result
Determinant branch	choose arg det branch	hidden selector unless certified	no primitive standing
CKM phase	use weak CP phase	separate ledger	no strong selector
θ -vacuum	relabel topological sector	representative change	no erasure
Vacuum energy profile	minimize $E(\bar{\theta})$ as if $\bar{\theta}$ were dynamical	response function, not selector	axion needed for dynamic minimization
Holonomy notation	rewrite sector data	notation unless residue-certified	no selector
Renormalization	scale/scheme movement	transport ledger	no endpoint fixation
EDM bound	use upper limit as derivation	calibration import	bound, not proof
Unspecified stabilizer	assert unknown mechanism for small nonzero $\bar{\theta}$	placeholder, not a certified source-continuity package	no reopening of $D_{\text{SCP},0}$; if specified, becomes $D_{\text{SCP},+}$ update
Massless quark	determinant-rank collapse / exact chiral freedom	singular physical-target change	not a selector inside fixed massive $D_{\text{SCP},0}$
PQ/axion	add dynamic field	richer-scope selector	legitimate if stable
UV CP/parity	add high-scale symmetry	richer-scope selector	legitimate if descent certified
Nelson-Barr	protect determinant phase	richer-scope selector	legitimate if radiatively stable
Flavor/modular	constrain determinant texture	richer-scope selector	legitimate if independently standing
Future EDM	positive empirical fixing record	scope update/witness	reconstruction required

B Claim and nonclaim register

ID	Claim	Status
SCP-T1	$\bar{\theta}$ is the unique strong-sector CP-odd quotient survivor.	proved from QCD anomaly/mass-phase ledger
SCP-T2	Generic nonzero $\bar{\theta}$ classes are exchanged with CP-conjugate classes; every nonzero class, including the sign-fixed π class, is non-identity topological load.	QCD-to-boundary and non-identity burden theorems
SCP-T3	Nonzero topological boundary load requires selector, input, witness, or richer descent.	AASC selector-burden theorem

ID	Claim	Status
SCP-T4	Same-scope chiral/anomaly/mass/CKM/vacuum/RG transports and endpoint-claim routes do not supply the selector burden.	transport and endpoint-claim exhaustion theorems
SCP-T5	$\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1$ and $\forall \kappa \neq 0, \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0$.	fixed-scope endpoint theorem
SCP-T6	In ordinary physics language, absent new physics the fixed strong-CP endpoint is $\bar{\theta} = 0$; formally, this means no selector, declared nonzero input, positive fixing witness, or richer-scope source-continuity package exists in $D_{\text{SCP},0}$.	physics-facing gloss of the endpoint theorem
SCP-N1	The θ -term is syntactically forbidden.	explicit nonclaim
SCP-N2	Parameterized nonzero-theta QCD cannot be studied.	explicit nonclaim
SCP-N3	Axion/PQ is invalid.	explicit nonclaim; classified as richer-scope selector
SCP-N4	A unique UV origin is selected.	explicit nonclaim
SCP-N5	Nonzero $\bar{\theta}$ is impossible in nature.	explicit nonclaim; a specified source-continuity package updates the target

C Forbidden-promotion register

Input	Forbidden promotion	Required certificate or endpoint
Written θ -coefficient	primitive nonzero endpoint	selector package, declared input, witness, or descent invariant quotient preserved
Chiral basis	erasure of $\bar{\theta}$	selector certificate needed
Anomaly equation	value oracle	independently standing branch rule
Mass determinant branch	determinant-phase fixation	vacuum-sector selector or representative collapse
θ -vacuum label	physical removal of CP load	dynamical carrier or selector law required
Vacuum-energy profile	same-scope relaxation of $\bar{\theta}$	empirical bound only; positive fixing witness
EDM upper bound	exact zero derivation	updates target
Axion model name	automatic solution	carrier, selection, quality, and descent certificate

Input	Forbidden promotion	Required certificate or endpoint
UV CP/parity label	automatic protection	breaking and radiative-stability certificate

D Anti-circularity audit

Table 11: Anti-circularity and forbidden-retyping audit.

Potential circle or misreading	Audit response
AASC is optional overlay.	Dependency inversion. Determinate endpoint evaluation already forces the kernel roles; AASC records the consequence layer.
$D_{\text{SCP},0}$ smuggles in $\bar{\theta} = 0$.	Proposition 0.2 states that the context fixes the slot, quotient, and candidate family only.
Nonzero C_κ is deleted by definition.	Deletion occurs only after quotient identification, free-topological-phase classification, non-identity load theorem, selector-burden theorem, same-scope transport audit, and endpoint-claim audit.
C_0 is chosen coefficient zero.	Proposition 6.12 states that identity selection is not $\theta = 0$, not $\arg \det M_q = 0$, and not displayed coefficient choice.
QCD permits θ , so nonzero C_κ stands.	Availability is not primitive endpoint standing; the theta coordinate slot is not a declared value.
EDM bounds select zero.	Upper-bound/null-channel records are constraints, not positive fixing witnesses and not exact selectors.
Vacuum energy selects zero.	$E(\bar{\theta})$ is a parameter-response profile unless a dynamical coordinate, such as an axion, is added.
Axions refute the theorem.	Axion/PQ is a richer-scope selector package, exactly as the theorem predicts.
CKM CP sources nonzero $\bar{\theta}$.	Typed source-continuity requires a certified determinant/descent bridge from weak CP to the strong topological boundary ledger.
$\theta = \pi$ escapes CP-odd deletion.	The π class is sign-fixed but non-identity topological load and therefore not the quotient identity.
The theorem bans all topological terms.	The theorem applies to freely variable strong-CP topological phase candidates lacking consistency/source fixation, not to quantized or consistency-fixed topological terms.
The theorem is only taxonomy.	Exhaustive status classification is the proof method; endpoint selection follows from certified survivor closure under Criterion 2.10.

Potential circle or misreading	Audit response
An unknown small- $\bar{\theta}$ stabilizer might exist.	If unspecified, it performs no standing-bearing work. If specified, it is a selector, input, positive witness, or descent package and updates the target to $D_{\text{SCP},+}$.

E Theorem-level source dependency map

The dependency entries below use the most stable identifiers available in the current project archive: theorem labels where the present manuscript supplies the object, source theorem or lemma titles where the source paper has stable numbering, and corpus-object labels where the source numbering is not stable. Title-only reference is avoided for load-bearing imports unless no narrower source identifier is available in the archived source package.

Strong-CP theorem or proof block	Imported source result object	Function in this manuscript
Dependency-inversion exclusion, Theorem 1.1; endpoint-use kernel theorem, Theorem 2.3	<i>Kernel</i> : target adequacy to <i>Adm</i> , <i>Stand</i> , <i>Ref</i> , <i>Irr</i> ; <i>Structure</i> : <i>AMetric</i> /pre-boundary and bivalent-admissible-interior objects; <i>P=NP</i> endpoint paper, mathematical-status and dependency-inversion posture only	Fixes reference, standing, admissibility, and irreversibility before candidate C_κ branches are evaluated; treats AASC as forced consequence layer rather than optional overlay.
Context non-smuggling, Proposition 0.2; proof-calculus audit, Proposition 2.6	<i>P=NP</i> endpoint paper, context non-smuggling and no-one-step-retyping posture; <i>Kernel</i> target-adequacy role discipline	Blocks readings in which $D_{\text{SCP},0}$ asserts C_0 by definition or deletes nonzero classes before proof.
Boundary non-promotion, Principle 2.11	EX6 Endpoint/Boundary No-Promotion object; EX1 Unresolved-Freedom Exclusion object; EX2 Quotient-Singleton Collapse object; EX3 Transport-Neutralization object; EX5 Calibration-Removal object	Prevents writable boundary traces, calibration records, quotient representatives, and transport artifacts from becoming primitive endpoint standing.
Quotient survivor, Theorem 3.1	<i>Structura</i> , Lemma “Strong CP: unique invariant $\bar{\theta}$ ”; <i>Standalone SM</i> , full renormalizable SM/QCD scope ledger	Identifies $\bar{\theta} = [\theta + \arg \det M_q]_{2\pi}$ as the audited object rather than either representative.
Topological bridge, Theorem 4.1	Gauge-Vacuum Compatibility, no-vacuum-anomaly-mask object; Residual Vacuum-Sector Rigidity theorem; EX7 Residual/Holonomy Exactness object; Curvature-Holonomy Obstruction theorem	Types $\bar{\theta}$ as topological-sector boundary weighting after quotienting local presentation.
Transport normal form, Theorem 5.1; endpoint-claim normal form, Theorem 5.2	A Fixed-Domain Exhaustion Theorem, same-domain comparison-class theorem; Closure by Exhaustion for Same-Scope Operators, closure-by-exhaustion object	Separates same-scope transport ledgers from endpoint-claim statuses and reduces both to audited normal forms.
Vacuum-energy non-selector, Lemma 5.9	Standard QCD vacuum-energy response over θ sectors plus AASC selector-burden and axion/richer-scope descent distinction	Blocks parameter-response minimization from acting as a dynamical same-scope selector.
Anomaly/Jacobian and determinant lemmas	Anomaly as Standing-Failure Obstruction theorem; Anomaly Ledger Algebraization, no-hidden-measure-or-determinant-selector object; Anomaly-Ledger Rigidity theorem; Mass-Matrix determinant/branch discipline object	Blocks anomaly equations, measure shifts, determinant branches, and mass-basis choices from becoming value selectors.

Strong-CP theorem or proof block	Imported source result object	Function in this manuscript
CKM firewall and weak-radiative lemma	Mixing-Angle and Phase Standing, CP-phase firewall object; Mass-Matrix determinant-descent discipline object	Separates weak CP standing from strong topological boundary selection unless a certified typed determinant/descent map is supplied.
Vacuum, vacuum-energy, and holonomy lemmas	Gauge-Vacuum Compatibility theorem; Residual Vacuum-Sector Rigidity theorem; Vacuum-Compatible Residual-Sector Normal Form; EX7 Wilson-loop/residue/anomaly-equation non-promotion objects; Wilson/holonomy notation nonpromotion object	Blocks θ -vacuum representative choice, nondynamical vacuum-energy response profiles, Wilson notation, holonomy residue, and local anomaly equations from becoming endpoint selectors.
Source-continuity chain, Lemmas 6.1–6.3 and Theorem 6.4	Kernel roles; EX6 boundary nonpromotion; EX3 transport-neutralization; EX1 hidden-selector and unresolved-freedom exclusions	Derives the source requirement for non-neutral CP-odd boundary load rather than stipulating it.
Selector-burden theorem, Theorem 6.8	Kernel role closure plus EX6 no-promotion, anomaly-ledger no-hidden-selector exclusions, fixed-domain closure, and the non-identity load theorem	Converts nonzero C_κ standing into an explicit selector/input/witness/descent burden.
Freely variable and non-identity burden, Definition 0.7, Theorem 6.5, and Lemma 6.7	EX6 Boundary Exactness object plus EX7 Holonomy/Residual Exactness object and Residual-Sector Rigidity theorem	Defines the freely variable strong-CP phase class and uniformly deletes every non-identity candidate without selector, including C_π , without overbroadly applying to consistency-fixed topological terms.
Identity availability, primitive standing, and non-coefficient-zero lock, Propositions 6.10–6.12	EX2 Quotient-Singleton discipline and EX6 Boundary-Load Absence object	Certifies the positive selection conditions for C_0 under Criterion 2.10 and separates identity selection from a chosen representative zero.
Fixed endpoint, Theorem 6.14	All preceding strong-CP theorem blocks	Concludes $\text{Sel}_{D_{\text{SCP},0}}(C_0) = 1$ and $\forall \kappa \neq 0, \text{Surv}_{D_{\text{SCP},0}}^{\text{cert}}(C_\kappa) = 0$.
Same-target closure, Theorems 9.5–9.6	AASC weakening-resistance and faithful counterexample architecture; neutrino endpoint paper proof-class comparator, Sections on exact-character selector burden and same-target closure	Specifies what a same-regime counterexample must preserve and which load-bearing link it must break.

F Notation and glossary

$D_{\text{SCP},0}$. Fixed AASC strong-CP endpoint target.

$D_{\text{SCP},+}$. Updated target after selector package, declared input, positive empirical fixing witness, or richer-scope descent.

$\bar{\theta}$. Physical strong-CP quotient survivor.

C_κ . Candidate endpoint class $\{\bar{\theta} = \kappa\}$, with C_0 the identity endpoint class.

Standing. Eligibility for use as part of the admitted ledger of a fixed domain.

Skin. Presentation, notation, branch, basis, or coordinate information that does not change admitted standing.

Selector package. Certified carrier, selection law, transport stability, and low-energy descent for nonzero $\bar{\theta}$.

Declared input. A boundary datum intentionally included as external data in the target theory.

Boundary load. Nontrivial endpoint or boundary information that affects the response ledger.

Positive fixing witness. Empirical CP-odd record $\text{Wit}_{\text{CP}}^+(C_\kappa)$ supporting a nonzero class or forcing a target update.

Bound record. Upper-bound or null-channel constraint $\text{Bound}_{\text{CP}}(\bar{\theta})$; not an endpoint selector.

Theta coordinate slot. The admitted quotient value space $\Theta_{\text{slot}} = \mathbb{R}/2\pi\mathbb{Z}$, distinct from a declared coefficient value.

Freely variable phase. A candidate C_κ not fixed by consistency theorem, quantization law, selector, declared input, positive fixing witness, or richer-scope descent.

Non-promotion. Failure of formal or boundary-relative data to become primitive standing absent a certificate.

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